HTML Quick-Styler

AI-Powered HTML Page Generator with IBM Granite 3.3 2B

Title: HTML Quick-Styler

Team members:

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Project Description:

HTML Quick-Styler is an intelligent web page generation platform that leverages IBM's Granite 3.3 2B Instruct model to provide comprehensive HTML/CSS page creation with professional styling and responsive design. The platform addresses the challenge of rapid web development by delivering AI-powered HTML generation, automatic section detection, modern CSS styling, color palette management, and production-ready code output.

Using Granite 3.3 2B's advanced language capabilities, the system analyzes user descriptions (page title, section requirements, color preferences), generates semantic HTML structure, applies modern CSS styling with animations, creates responsive layouts, and provides multiple color palette options. The system ensures professional output through Gradio's user-friendly interface while maintaining high-quality design standards through sophisticated template generation and AI-powered content suggestions.

HTML Quick-Styler transforms web development into an intelligent, user-friendly experience through its modern Gradio interface, comprehensive feature set, and AI-powered analysis that provides personalized HTML pages considering design trends, accessibility standards, and responsive design principles.

Scenarios:

Scenario 1: Portfolio Website Template

A designer or freelancer wants to quickly build a professional online portfolio to showcase their work and skills. The system provides a ready-made Portfolio Website template that includes essential sections such as Hero, Portfolio Gallery, About, Contact, and Footer. Each section is structured with a clean layout to highlight projects and personal achievements effectively. The template allows users to easily replace images, descriptions, and links with their own content. It ensures a visually balanced design while maintaining consistency across all sections. Additionally, the responsive structure ensures that the portfolio displays perfectly on desktops, tablets, and mobile devices.

Scenario 2: SaaS Landing Page Template

A startup or software company needs a compelling landing page to launch their new SaaS product and convert visitors into customers. The system provides a ready-made SaaS Landing Page template that includes essential sections such as Hero Banner, Features Showcase, Pricing Plans Comparison, Testimonials, Call-to-Action Button, and Footer. Each section is strategically designed to highlight product benefits, build trust through social proof, and drive conversions. The template allows businesses to easily customize product names, pricing tiers, and feature descriptions. The conversion-focused design with strategic placement of CTA buttons ensures maximum user engagement and sign-ups across all devices.

Scenario 3: Company Website Template

A small to medium-sized business wants to establish a professional online presence that builds credibility and attracts potential clients. The system provides a ready-made Company Website template that includes essential sections such as Header with Navigation, Hero Section, About Company, Services List, Team Members, Contact Form, and Footer. Each section is designed to communicate company values, showcase expertise, and build trust with visitors. The template allows businesses to easily update team photos, service offerings, and company descriptions. The professional layout ensures that visitors can quickly understand the company's offerings and easily get in touch across all devices.

Scenario 4: E-Commerce Store Template

An online retailer needs an attractive and functional storefront to showcase products and drive sales. The system provides a ready-made E-Commerce Store template that includes essential sections such as Hero Banner with Sale

Announcement, Product Grid with 4 Items, Featured Categories, Shopping Features, Newsletter Signup, and Footer. Each section is optimized to highlight products, encourage browsing, and capture leads. The template allows store owners to easily update product images, prices, and category descriptions. The shopping-focused design with clear product displays and newsletter integration helps maximize conversions and build customer relationships across all devices.

Scenario 5: Personal Blog Template

A writer, content creator, or influencer wants to share their thoughts, stories, and expertise with a growing audience. The system provides a ready-made Personal Blog template that includes essential sections such as Hero with Author Intro, Blog Posts Grid with 3 Articles, Categories Sidebar, Popular Posts, Newsletter, and Footer. Each section is crafted to highlight content, improve readability, and encourage engagement. The template allows bloggers to easily update post titles, featured images, and category listings. The content-first design with clean typography and strategic newsletter placement helps build a loyal readership across all devices.

Scenario 6: Restaurant Website Template

A restaurant owner needs an appetizing online presence that showcases their menu, atmosphere, and attracts diners. The system provides a ready-made Restaurant Website template that includes essential sections such as Hero with Food Image, Menu Section with Specials, About Chef, Reservation Form, Customer Reviews, Location Map, and Footer. Each section is designed to tantalize taste buds, build credibility, and make reservations easy. The template allows restaurant owners to easily update menu items, chef bios, and business hours. The hunger-inducing design with prominent reservation options ensures a steady stream of customers across all devices.

Scenario 7: Creative Agency Template

A design, marketing, or creative agency wants to showcase their portfolio of work and attract high-value clients. The system provides a ready-made Creative Agency template that includes essential sections such as Hero with Video Background, Services Offered, Case Studies Gallery, Team Members, Client Logos, Contact Section, and Footer. Each section is designed to demonstrate creative capabilities, build trust through social proof, and convert visitors into clients. The template allows agencies to easily update project case studies, team photos, and service descriptions. The portfolio-focused design with impressive case study displays helps win new business across all devices.

Scenario 8: App Landing Page Template

A mobile app developer needs a dedicated page to promote their app, highlight features, and drive downloads. The system provides a ready-made App Landing Page template that includes essential sections such as Hero with App Screenshot, Features with Icons, How It Works Steps, Download Buttons, User Reviews, FAQ, and Footer. Each section is optimized to showcase app functionality, address user questions, and encourage installation. The template allows developers to easily update app screenshots, feature descriptions, and FAQ answers. The conversion-optimized design with prominent app store buttons ensures maximum downloads across all devices.

Scenario 9: Fitness Center Template

A gym, fitness studio, or personal trainer needs an energetic website that attracts new members and showcases their facilities. The system provides a ready-made Fitness Center template that includes essential sections such as Hero with Class Schedule, Trainer Profiles, Membership Plans, Facility Gallery, Client Transformations, Contact, and Footer. Each section is designed to motivate visitors, build trust in trainers, and make membership sign-up easy. The template allows fitness businesses to easily update class times, trainer bios, and membership pricing. The energetic design with transform

Architecture Overview

HTML Quick-Styler is built as a modular platform combining Python backend with IBM Granite 3.3 2B Instruct model for intelligent HTML generation. The architecture prioritizes user experience, design quality, and code production through Gradio's interactive interface and sophisticated template generation.

Core Technologies

- Python 3.x: Core programming language for application logic
- IBM Granite 3.3 2B Instruct: AI model for intelligent content generation
- Gradio: Interactive web interface framework for user interaction
- HTML5/CSS3: Modern web standards for page generation
- JavaScript: Client-side interactivity and animations
- Google Colab: Cloud-based development environment
- Responsive Design: Mobile-first approach with media queries

Pre-requisites

Software Requirements

- Python 3.8+: Download from python.org
- Google Account: Required for Google Colab access
- Gradio: Install via pip (pip install gradio)
- Web Browser: Chrome, Firefox, Safari, or Edge
- Internet Connection: Required for Colab and model access

Knowledge Prerequisites

- Python Basics: Functions, classes, dictionaries, string manipulation
- HTML/CSS: Understanding of HTML structure and CSS styling
- Gradio Framework: Basic understanding of interface components

- Web Design: Color theory, responsive design principles
- Google Colab: Familiarity with notebook interface

Hardware Requirements

- Processor: Intel i3 or equivalent (Colab runs on cloud)
- RAM: Minimum 4GB local (Colab provides GPU/TPU)
- Storage: 1GB free space for generated files
- Display: Minimum 1280x720 resolution for interface

Project Workflow

Phase 1: Environment Setup & Model Configuration

Establish Google Colab environment and configure Granite model

Activity 1.1: Set up Google Colab Environment

- Create new Google Colab notebook
- Enable GPU runtime for better performance
- Install required Python packages
- Verify environment configuration

Activity 1.2: Configure IBM Granite Model

- Access Hugging Face model repository
- Load Granite 3.3 2B Instruct model
- Test model inference capabilities
- Configure generation parameters

Activity 1.3: Install and Configure Gradio

- Install Gradio via pip
- Test basic Gradio interface
- Configure sharing options
- Verify interface rendering

Phase 2: Core Application Development

Build HTML generation engine and template system

Activity 2.1: Build Color Palette Generator

- Create ColorPaletteGenerator class
- Define 8 color palette schemes
- Implement palette selection logic

- Add color validation methods

Activity 2.2: Implement HTML Page Builder

- Create HTMLPageBuilder class
- Build section detection algorithm
- Implement template generation methods
- Create SEO meta tags generator

Activity 2.3: Develop CSS Generation Engine

- Build comprehensive CSS generator
- Implement responsive design rules
- Add animation and transition effects
- Create mobile-first media queries

Activity 2.4: Build Section Templates

- Create hero section template
- Build features grid template
- Implement portfolio gallery template
- Add contact form template
- Create pricing and testimonial templates

Phase 3: Gradio Interface Development

Create interactive user interface with multiple tabs

Activity 3.1: Interactive HTML Generation & Preview Interface

Activity 3 focuses on designing and implementing a **complete HTML generation and visualization interface** that enables users to create, customize, preview, and understand web page layouts through an intuitive, multi-section workflow.

Phase 4: Google Colab Deployment:

Integrate all components and perform comprehensive testing

Activity 4.1: End-to-End Integration

- Connect all interface components
- Test data flow between tabs
- Validate generation pipeline
- Test error handling

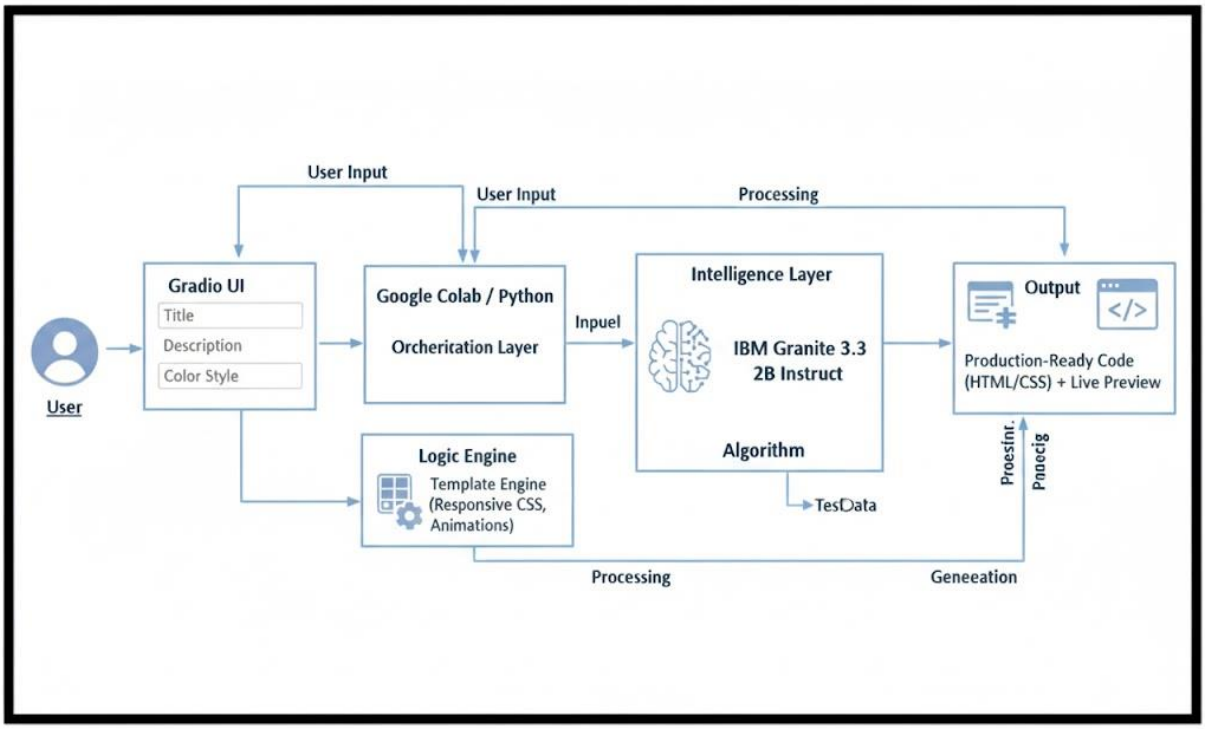
Phase 5: Deployment & Documentation

Deploy application and create comprehensive documentation

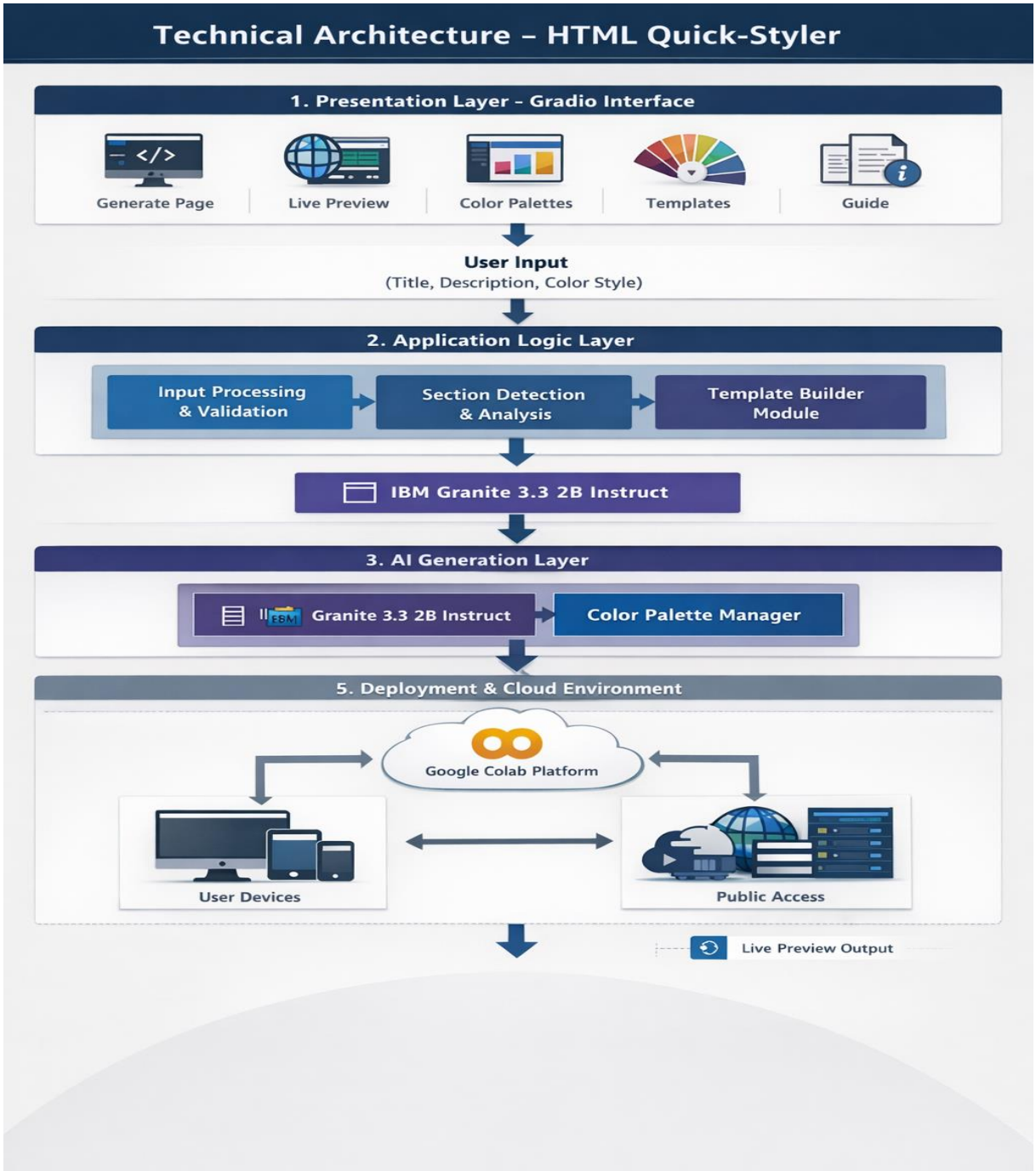
Activity 5.1: Functionality testing

- Configure Gradio sharing settings
- Enable public access link
- Test external access
- Document deployment steps

Technical Architecture:



Project Structure:



MILESTONE 1: Environment Setup & Model Configuration

Establish Google Colab environment and configure Granite model

Activity 1.1: Install Dependencies & Setup Environment

Create requirements.txt:

```
!pip install -q gradio

import gradio as gr
import json
import re
from datetime import datetime
from typing import Tuple, Dict, List
import random
```

Activity 1.2: Configure IBM Granite Model

Access Hugging Face and load model:

Model: ibm-granite/granite-3.3-2b-instruct

Colab Notebook:

MILESTONE 2: Core Application Development

Activity 2.1: Color Palette Generator Implementation

Create ColorPaletteGenerator class with 8 predefined color schemes:

```
class ColorPaletteGenerator:
    """Generate visually cohesive color palettes"""

    @staticmethod
    def generate_palette(style: str = "modern") -> Dict[str, str]:
        """Generate a cohesive color palette based on style"""

        palettes = {
            'modern': {
                'primary': '#2c3e50',
                'secondary': '#34495e',
                'accent': '#3498db',
                'background': '#0f172a',
                'surface': '#1e293b',
                'text_primary': '#f1f5f9',
                'text_secondary': '#cbd5e1',
                'border': '#334155'
```

```
        },
        'vibrant': {
            'primary': '#27ae60',
            'secondary': '#1e8449',
            'accent': '#52be80',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'ocean': {
            'primary': '#0099cc',
            'secondary': '#006699',
            'accent': '#00ccff',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'forest': {
            'primary': '#2ecc71',
            'secondary': '#27ae60',
            'accent': '#f1c40f',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        '


          'sunset': {
            'primary': '#ff8c42',
            'secondary': '#ff6b35',
            'accent': '#ffa751',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'luxury': {
            'primary': '#1a1a2e',
            'secondary': '#16213e',
            'accent': '#e94560',
            'background': '#0a0a0f',
            'surface': '#1e1e2e',
            'text_primary': '#f1f5f9',
            'text_secondary': '#a0a0b0',
            'border': '#2a2a3a'
    },
        'creative': {
```

```
            'primary': '#667eea',
            'secondary': '#764ba2',
            'accent': '#f093fb',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'minimal': {
            'primary': '#9b59b6',
            'secondary': '#8e44ad',
            'accent': '#e74c3c',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        }
    }

    return palettes.get(style.lower(), palettes['modern'])
```

Activity 2.2: HTML Page Builder Implementation

The HTMLPageBuilder class includes:

- Section extraction from user descriptions
- Keyword detection for hero, features, about, portfolio, testimonials, pricing, contact, and footer sections
- SEO meta tags generation with Open Graph and Twitter Card support
- Individual section HTML generation with professional templates

```
@staticmethod
 def generate_section_html(section_type: str, palette: Dict) -> str:
    """Generate HTML for specific section with proper containers"""
 sections = {
        'hero': f'''        <section class="hero-section">
        <div class="container">
            <div class="hero-content">
                <h1>Welcome to Your Site</h1>
                <p>Create amazing web experiences</p>
                <button class="btn btn-primary">Get Started</button>
            </div>
        </div>
    </section>''',
          'features': f'''        <section class="features-section">
        <div class="container">
            <h2 class="section-title">Features</h2>
```

```
            <div class="features-grid">
              <div class="feature-card">
                <div class="feature-icon">⚡</div>
                <h3>Fast</h3>
                <p>Lightning-quick performance</p>
              </div>
              <div class="feature-card">
                <div class="feature-icon">🔒</div>
                <h3>Secure</h3>
                <p>Enterprise-grade security</p>
              </div>
              <div class="feature-card">
                <div class="feature-icon">📈</div>
                <h3>Scalable</h3>
                <p>Grow without limits</p>
              </div>
            </div>
          </div>
        </section>''',
            'about': f'''       <section class="about-section">
          <div class="container">
            <div class="about-content">
              <h2 class="section-title">About Us</h2>
              <p>We're passionate about creating beautiful, functional websites that help your
business grow.</p>
            </div>
          </div>
        </section>''',
            'portfolio': f'''       <section class="portfolio-section">
          <div class="container">
            <h2 class="section-title">Our Work</h2>
            <div class="portfolio-grid">
              <div class="portfolio-item">
                <div class="portfolio-overlay">
                  <h4>Project 1</h4>
                  <p>Web Design</p>
                </div>
              </div>
              <div class="portfolio-item">
                <div class="portfolio-overlay">
                  <h4>Project 2</h4>
                  <p>Mobile App</p
            </div>
              </div>
              <div class="portfolio-item">
                <div class="portfolio-overlay">
                  <h4>Project 3</h4>
                  <p>Branding</p>
                </div>
              </div>
            </div>
          </div>
        </section>''',
```

```
        'testimonials': f'''        <section class="testimonials-section">
        <div class="container">
            <h2 class="section-title">What People Say</h2>
            <div class="testimonials-grid">
                <div class="testimonial-card">
                    <p>"Amazing service! They delivered beyond our expectations."</p>
                    <div class="testimonial-author">
                        <strong>- Sarah Johnson</strong>
                        <span>CEO, TechStart</span>
                    </div>
                </div>
                <div class="testimonial-card">
                    <p>"Highly recommended! Professional and creative team."</p>
                    <div class="testimonial-author">
                        <strong>- Michael Chen</strong>
                        <span>Founder, DesignLab</span>
                    </div>
                </div>
            </div>
        </div>
    </section>''',

        'pricing': f'''        <section class="pricing-section">
        <div class="container">
            <h2 class="section-title">Pricing Plans</h2>
            <div class="pricing-grid">
                <div class="pricing-card">
<h3>Starter</h3>
                    <p class="price">$29</p>
                    <ul>
                        <li>✓ Feature 1</li>
                        <li>✓ Feature 2</li>
                        <li>✓ Feature 3</li>
                    </ul>
                    <button class="btn btn-secondary">Get Started</button>
                </div>
                <div class="pricing-card featured">
                    <h3>Pro</h3>
                    <p class="price">$79</p>
                    <ul>
                        <li>✓ All Starter features</li>
                        <li>✓ Feature 4</li>
                        <li>✓ Feature 5</li>
                    </ul>
                    <button class="btn btn-primary">Get Started</button>
                </div>
            </div>
        </div>
    </section>''',

        'contact': f'''        <section class="contact-section">
        <div class="container">
```

```
        <h2 class="section-title">Get in Touch</h2>
        <div class="contact-wrapper">
          <form class="contact-form">
            <div class="form-group">
              <input type="text" placeholder="Your Name" required>
            </div>
            <div class="form-group">
              <input type="email" placeholder="Your Email" required>
            </div>
            <div class="form-group">
              <textarea placeholder="Your Message" rows="5" required></textarea>
            </div>
            <button type="submit" class="btn btn-primary">Send Message</button>
          </form>
        </div>
      </div>
    </section>''',

      'footer': f''       <footer class="footer">
      <div class="container">
        <div class="footer-content">
          <p>&copy; 2024 Your Company. All rights reserved.</p>
          <div class="footer-links">
            <a href="#privacy">Privacy</a>
            <a href="#terms">Terms</a>
            <a href="#contact">Contact</a>
          </div>
        </div>
      </div>
    </footer>'''
    }
    return sections.get(section_type, '')
```

Activity 2.3: CSS Generation Engine

The CSS generator creates comprehensive styling including:

- CSS variables for dynamic color management
- Gradient backgrounds and modern animations
- Responsive grid layouts with Flexbox and CSS Grid
- Smooth transitions and hover effects
- Mobile-first media queries for tablets and phones
- Sticky header with backdrop blur effect
- Card-based components with depth and shadows

```
@staticmethod
    def generate_css(palette: Dict) -> str:
        """Generate comprehensive CSS with proper containers"""
        css = f'''    <style>
        @import
url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=
Playfair+Display:wght@400;500;600;700;800&display=swap');

        * {{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }}

        :root {{
            --primary: {palette['primary']};
            --secondary: {palette['secondary']};
            --accent: {palette['accent']};
            --background: {palette['background']};
            --surface: {palette['surface']};
            --text-primary: {palette['text_primary']};
            --text-secondary: {palette['text_secondary']};
            --border: {palette['border']};
            --font-primary: 'Inter', sans-serif;
            --font-heading: 'Playfair Display', serif;
        }}

        body {{
            font-family: var(--font-primary);
            background: linear-gradient(135deg, var(--background) 0%, var(--surface) 100%);
            color: var(--text-primary);
            line-height: 1.6;
        }}

        /* Container for all sections */
        .container {{
            max-width: 1200px;
            margin: 0 auto;
            padding: 0 20px;
        }}

        /* Section styling */
        section {{
            padding: 80px 0;
            position: relative;
        }}

        .section-title {{
            font-family: var(--font-heading);
            font-size: 2.5rem;
            text-align: center;
            margin-bottom: 50px;
            color: var(--text-primary);
            position: relative;
```

```
}}

.section-title::after {{
    content: '';
    position: absolute;
    bottom: -15px;
    left: 50%;
    transform: translateX(-50%);
    width: 80px;
    height: 4px;
    background: linear-gradient(90deg, var(--primary), var(--accent));
    border-radius: 2px;
}}

/* Header */
header {{
    background: rgba(15, 23, 42, 0.95);
    backdrop-filter: blur(10px);
    padding: 1.5rem 0;
    position: sticky;
    top: 0;
    z-index: 100;
    border-bottom: 1px solid var(--border);
}}

header .container {{
    display: flex;
    justify-content: space-between;
    align-items: center;
}}
header h1 {{
    font-size: 1.8rem;
    font-weight: 700;
    background: linear-gradient(135deg, var(--primary), var(--accent));
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    background-clip: text;
}}

header nav a {{
    color: var(--text-secondary);
    text-decoration: none;
    margin-left: 2rem;
    transition: color 0.3s ease;
    font-weight: 500;
}}

header nav a:hover {{
    color: var(--primary);
}}

/* Hero Section */
.hero-section {{
    background: linear-gradient(135deg, var(--primary) 0%, var(--secondary) 100%);
```

```
    color: white;
    min-height: 600px;
    display: flex;
    align-items: center;
    text-align: center;
}}

.hero-content h1 {{
    font-size: 4rem;
    margin-bottom: 1rem;
    font-family: var(--font-heading);
    color: white;
}}

.hero-content p {{
    font-size: 1.35rem;
    margin-bottom: 2rem;
    color: rgba(255, 255, 255, 0.9);
    max-width: 600px;
    margin-left: auto;
    margin-right: auto;
}}

/* Buttons */
.btn {{
    padding: 12px 30px;
    border: none;
    border-radius: 8px;
    font-size: 1rem;
    cursor: pointer;
    transition: all 0.3s ease;
    font-weight: 600;
    display: inline-block;
    text-decoration: none;
}}

.btn-primary {{
    background: var(--primary);
    color: white;
    box-shadow: 0 4px 15px rgba(0,0,0,0.3);
}}

.btn-primary:hover {{
    transform: translateY(-2px);
    box-shadow: 0 6px 20px rgba(0,0,0,0.4);
    background: var(--accent);
}}

.btn-secondary {{
    background: var(--secondary);
    color: white;
}}

.btn-secondary:hover {{
```

```
    background: var(--primary);
    transform: translateY(-2px);
}}

/* Features Section */
.features-section {{
    background: var(--surface);
}}

.features-grid {{
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 30px;
}}

.feature-card {{
    background: rgba(30, 41, 59, 0.8);
    padding: 40px 30px;
    border-radius: 16px;
    border: 1px solid var(--border);
    transition: all 0.3s ease;
    text-align: center;
    backdrop-filter: blur(10px);
}}

.feature-card:hover {{
    transform: translateY(-8px);
    border-color: var(--primary);
    box-shadow: 0 20px 40px rgba(0,0,0,0.3);
}}

.feature-icon {{
    font-size: 3rem;
    margin-bottom: 20px;
}}
.feature-card h3 {{
    color: var(--primary);
    margin-bottom: 15px;
    font-size: 1.5rem;
}}

/* About Section */
.about-section {{
    background: var(--background);
    text-align: center;
}}

.about-content {{
    max-width: 800px;
    margin: 0 auto;
    background: rgba(30, 41, 59, 0.5);
    padding: 50px;
    border-radius: 20px;
    border: 1px solid var(--border);
```

```
    backdrop-filter: blur(10px);
}}

.about-content p {{
    font-size: 1.2rem;
    color: var(--text-secondary);
    line-height: 1.8;
}}

/* Portfolio Section */
.portfolio-section {{
    background: var(--surface);
}}

.portfolio-grid {{
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(350px, 1fr));
    gap: 30px;
}}

.portfolio-item {{
    aspect-ratio: 1;
    background: linear-gradient(135deg, var(--primary), var(--secondary));
    border-radius: 16px;
    position: relative;
    overflow: hidden;
    cursor: pointer;
    border: 1px solid var(--border);
}}

.portfolio-overlay {{
    position: absolute;
    bottom: -100%;
    left: 0;
    right: 0;
    background: rgba(0, 0, 0, 0.8);
    padding: 20px;
    transition: bottom 0.3s ease;
    text-align: center;
    backdrop-filter: blur(5px);
}}

.portfolio-item:hover .portfolio-overlay {{
    bottom: 0;
}}

.portfolio-overlay h4 {{
    color: white;
    margin-bottom: 5px;
    font-size: 1.2rem;
}}

.portfolio-overlay p {{
    color: rgba(255, 255, 255, 0.8);
```

```
    font-size: 0.9rem;
}}

/* Testimonials Section */
.testimonials-section {{
    background: var(--background);
}}

.testimonials-grid {{
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 30px;
}}

.testimonial-card {{
    background: rgba(30, 41, 59, 0.8);
    padding: 30px;
    border-left: 4px solid var(--primary);
    border-radius: 12px;
    backdrop-filter: blur(10px);
    border: 1px solid var(--border);
}}

.testimonial-card p {{
    font-style: italic;
    margin-bottom: 20px;
    color: var(--text-secondary);
    font-size: 1.1rem;
    line-height: 1.6;
}}

.testimonial-author {{
    display: flex;
    flex-direction: column;
}}

.testimonial-author strong {{
    color: var(--primary);
    font-size: 1rem;
}}
.testimonial-author span {{
    color: var(--text-secondary);
    font-size: 0.9rem;
}}

/* Pricing Section */
.pricing-section {{
    background: var(--surface);
}}

.pricing-grid {{
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 30px; }}
```

```
.pricing-card {{
   background: rgba(30, 41, 59, 0.8);
   padding: 40px 30px;
   border-radius: 16px;
   border: 1px solid var(--border);
   text-align: center;
   transition: all 0.3s ease;
   backdrop-filter: blur(10px);
}}

.pricing-card.featured {{
   border-color: var(--primary);
   box-shadow: 0 20px 40px rgba(0,0,0,0.3);
   transform: scale(1.05);
}}

.pricing-card h3 {{
   font-size: 1.8rem;
   margin-bottom: 15px;
   color: var(--text-primary);
}}

.pricing-card .price {{
   font-size: 3rem;
   color: var(--primary);
   margin: 20px 0;
   font-weight: 700;
}}

.pricing-card ul {{
   list-style: none;
   margin: 30px 0;
   text-align: left;
}}

.pricing-card li {{
   padding: 10px 0;
   color: var(--text-secondary);
   border-bottom: 1px solid var(--border);
}}

.pricing-card li:last-child {{
   border-bottom: none;
}}

/* Contact Section */
.contact-section {{
   background: var(--background);
}}

.contact-wrapper {{
   max-width: 600px;
   margin: 0 auto;
```

```
    background: rgba(30, 41, 59, 0.8);
    padding: 40px;
    border-radius: 20px;
    border: 1px solid var(--border);
    backdrop-filter: blur(10px);
}}

.contact-form {{
    display: flex;
    flex-direction: column;
    gap: 20px;
}}

.form-group input,
.form-group textarea {{
    width: 100%;
    padding: 15px;
    border: 1px solid var(--border);
    border-radius: 8px;
    background: rgba(15, 23, 42, 0.8);
    color: var(--text-primary);
    font-family: var(--font-primary);
    font-size: 1rem;
}}

.form-group input:focus,
.form-group textarea:focus {{
    outline: none;
    border-color: var(--primary);
    box-shadow: 0 0 0 3px rgba(52,152,219,0.2);
}}

.form-group input::placeholder,
.form-group textarea::placeholder {{
    color: var(--text-secondary);
    opacity: 0.7;
}}

/* Footer */
.footer {{
    background: rgba(15, 23, 42, 0.95);
    border-top: 1px solid var(--border);
    padding: 40px 0;
}}

.footer-content {{
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 20px;
}}
.footer-links {{
    display: flex;
```

```
        gap: 30px;
    }}

    .footer-links a {{
        color: var(--text-secondary);
        text-decoration: none;
        transition: color 0.3s ease;
    }}

    .footer-links a:hover {{
        color: var(--primary);
    }}

    /* Responsive */
    @media (max-width: 768px) {{
        header .container {{
            flex-direction: column;
            gap: 15px;
        }}

        header nav a {{
            margin: 0 10px;
        }}

        .hero-content h1 {{
            font-size: 2.5rem;
        }}

        .section-title {{
            font-size: 2rem;
        }}

        .pricing-card.featured {{
            transform: scale(1);
        }}

        .footer-content {{
            flex-direction: column;
            text-align: center;
        }}

        .footer-links {{
            justify-content: center;
        }}
    }}
</style>'''
    return css
```

Activity 3.1: Main Interface with Multiple Tabs

The Gradio interface includes 5 tabs:

class HTMLGenerationEngine:

```
    """Main engine for HTML page generation"""
    @staticmethod
    def generate_page(title: str, description: str, style: str = "modern") -> Tuple[str, str]:
        """Generate complete HTML page"""
        # Generate color palette
        palette = ColorPaletteGenerator.generate_palette(style)
        # Extract sections
        sections = HTMLPageBuilder.extract_sections(description)
        # Generate meta tags
        meta_tags = HTMLPageBuilder.generate_meta_tags(title, description)
        # Generate sections
        sections_html = '\n\n'.join([
            HTMLPageBuilder.generate_section_html(section, palette)
            for section in sections
            if HTMLPageBuilder.generate_section_html(section, palette)
        ])
        # Generate CSS
        css = HTMLPageBuilder.generate_css(palette)
        # Build complete HTML
        html = f"<!DOCTYPE html>
<html lang="en">
<head>
{meta_tags}
{css}
</head>
<body>
    <header>
```

```
        <div class="container">
            <h1>{title}</h1>
            <nav>
                <a href="#features">Features</a>
                <a href="#about">About</a>
                <a href="#contact">Contact</a>
            </nav>
        </div>
    </header>
    <main>
{sections_html}
    </main>
</body>
</html>'''
        return html, palette
generation_history = []
def generate_html_page(title: str, description: str, style: str = "modern") -> Tuple[str, str, str]:
    """Generate HTML page with styling"""
    if not title.strip() or not description.strip():
        return "", "❌ Please provide both title and description!", ""
    try:
        html_output, palette = HTMLGenerationEngine.generate_page(title, description, style)
            # Get color preview HTML
        color_preview_html = ColorPaletteGenerator.get_color_preview_html(palette)
        return html_output, "✅ HTML page generated successfully!", color_preview_html
    except Exception as e:
        return "", f"❌ Error: {str(e)}", ""
```

```
def get_color_suggestions(style: str) -> str:

    """Get color suggestions based on style with visual preview"""

    palette = ColorPaletteGenerator.generate_palette(style)

     # Get color preview HTML

    color_preview_html = ColorPaletteGenerator.get_color_preview_html(palette)

      return color_preview_html

# Template functions

def load_portfolio():

    return ("Portfolio Website", "Hero section with welcome banner, portfolio gallery with 6 items, about section, testimonials, contact form, footer", "modern")

def load_saas():

    return ("SaaS Product", "Hero banner, features showcase with 3 items, pricing plans comparison, testimonials, call-to-action button, footer", "vibrant")

def load_company():

    return ("Company Website", "Header with navigation, hero section, about company, services list, team members, contact form, footer", "ocean")

def load_ecommerce():

    return ("E-Commerce Store", "Hero banner with sale, product grid with 4 items, featured categories, shopping features, newsletter signup, footer", "luxury")

def load_blog():

    return ("Personal Blog",  "Hero with author intro, blog posts grid with 3 articles, categories sidebar, popular posts, newsletter, footer", "creative")

def load_restaurant():

    return ("Restaurant Website", "Hero with food image, menu section with specials, about chef, reservation form, customer reviews, location map, footer", "sunset")

def load_agency():

    return ("Creative Agency", "Hero with video background, services offered, case studies gallery, team members, client logos, contact section, footer",  "creative")

def load_app_landing():

    return ("App Landing Page", "Hero with app screenshot, features with icons, how it works steps, download buttons, user reviews, FAQ, footer", "vibrant")
```

```
def load_fitness():

    return ("Fitness Center", "Hero with class schedule, trainer profiles, membership plans, facility gallery, client transformations, contact, footer", "forest")

print("🚀 Loading HTML Quick-Styler...")

# Custom CSS for Gradio interface

gradio_css = """

@import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&display=swap');

.gradio-container, .gradio-container * {

    font-family: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif!important;

}

.tabs {

    border: none !important;

    background: transparent !important;

}

.tab-nav {

    display: flex !important;

    gap: 4px !important;

    border-bottom: 1px solid #334155 !important;

    padding-bottom: 8px !important;

    margin-bottom: 20px !important;

}

.tab-nav button {

    background: transparent !important;

    border: none !important;

    padding: 10px 20px !important;

    color: #94a3b8 !important;
```

```
    font-weight: 500 !important;
    font-size: 14px !important;
    border-radius: 8px !important;
    transition: all 0.2s ease !important;
}
.tab-nav button:hover {
    background: #1e293b !important;
    color: #f1f5f9 !important;
}
.tab-nav button.selected {
    background: #2d3a4f !important;
    color: #fff !important;
    font-weight: 600 !important;
}
.gr-box, .gr-form, .gr-panel {
    border: 1px solid #334155 !important;
    border-radius: 12px !important;
    background: #1a1a2e !important;
    padding: 20px !important;
    box-shadow: 0 4px 20px rgba(0,0,0,0.2) !important;
}
input, textarea, select, .gr-input, .gr-dropdown {
    background: #0f172a !important;
    border: 1px solid #334155 !important;
    border-radius: 8px !important;
    color: #f1f5f9 !important;
    padding: 12px !important;
```

```
    font-size: 14px !important;
    transition: all 0.2s ease !important;
}
input:focus, textarea:focus, select:focus {
    border-color: #3498db !important;
    box-shadow: 0 0 0 2px rgba(52,152,219,0.2) !important;
    outline: none !important;
}
label {
    font-size: 12px !important;
    font-weight: 600 !important;
    color: #94a3b8 !important;
    text-transform: uppercase !important;
    letter-spacing: 0.3px !important;
    margin-bottom: 6px !important;
}
.gr-button, button {
    border-radius: 8px !important;
    font-weight: 600 !important;
    font-size: 14px !important;
    padding: 10px 20px !important;
    transition: all 0.2s ease !important;
    border: none !important;
    cursor: pointer !important;
}
.gr-button-primary {
background: #3498db !important;
```

```
    color: white !important;
}
.gr-button-primary:hover {
    background: #2980b9 !important;
    transform: translateY(-1px) !important;
    box-shadow: 0 4px 12px rgba(52,152,219,0.3) !important;
}
.gr-button-secondary {
    background: #2d3a4f !important;
    color: #f1f5f9 !important;
}
.gr-button-secondary:hover {
    background: #3a4a63 !important;
}
.template-grid {
    display: grid !important;
    grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)) !important;
    gap: 16px !important;
    margin: 20px 0 !important;
}
.template-card {
    background: #1e1e32 !important;
    border: 1px solid #2d3a4f !important;
    border-radius: 12px !important;
    padding: 20px !important;
    transition: all 0.2s ease !important;
    height: 100% !important;
```

```
    display: flex !important;
    flex-direction: column !important;
}
.template-card:hover {
    border-color: #3498db !important;
    transform: translateY(-2px) !important;
    box-shadow: 0 8px 24px rgba(0,0,0,0.3) !important;
}
.template-card h4 {
    color: #f1f5f9 !important;
    font-size: 16px !important;
    font-weight: 600 !important;
    margin-bottom: 8px !important;
}
.template-card p {
    color: #94a3b8 !important;
    font-size: 13px !important;
    margin-bottom: 16px !important;
    flex-grow: 1 !important;
    line-height: 1.5 !important;
}
.template-card .gr-button {
    margin-top: auto !important;
    background: #2d3a4f !important;
    color: #f1f5f9 !important;
    font-size: 12px !important;
    padding: 8px !important;
```

```
}
.template-card .gr-button:hover {
    background: #3498db !important;
}
"""
with gr.Blocks(title="HTML Quick-Styler", css=gradio_css) as demo:
    gr.Markdown("""
    # 🎨 HTML Quick-Styler
    Generate professional HTML pages with automatic styling
    """)
    with gr.Tabs():
```

Tab 1: Generate Page

```
 with gr.TabItem("Generator"):
            with gr.Row():
                with gr.Column(scale=1):
                    gr.Markdown("### Page Details")
                    page_title = gr.Textbox(
                        label="Title",
                        value="Modern Portfolio Website",
                        lines=1
                    )
                    page_description = gr.Textbox(
label="Description",
                        value="Hero section with welcome message, features list with 3 items, about section, portfolio gallery, testimonials, contact form, and footer",
                        lines=4
                    )
```

```
            color_style = gr.Dropdown(
                ["modern", "vibrant", "ocean", "forest", "sunset", "luxury", "creative",
"minimal"],
                value="modern",
                label="Color Style"
            )
            generate_btn = gr.Button("Generate HTML", size="lg", variant="primary")
        with gr.Column(scale=1):
            gr.Markdown("### Output")
            html_output = gr.Code(
                language="html",
                label="HTML Code",
                lines=20
            )
            status_msg = gr.Markdown()
            color_info = gr.HTML()
    generate_btn.click(
        generate_html_page,
        inputs=[page_title, page_description, color_style],
        outputs=[html_output, status_msg, color_info]
    )
```

Tab 2: Live Preview

```
 with gr.TabItem("Preview"):
    gr.Markdown("### Live Preview")
    with gr.Row():
        preview_button = gr.Button("Load Preview", size="lg", variant="primary")
        refresh_preview = gr.Button("Refresh", size="lg", variant="secondary")
```

```
        preview_output = gr.HTML(value="<p style='text-align:center; color:#fff; padding:40px;'>No preview loaded</p>")

        def show_preview(html_text):

            if not html_text or not html_text.strip():

                return "<p style='text-align:center; color:#fff; padding:40px;'>No HTML provided</p>"

            return html_text

        preview_button.click(

            show_preview,

            inputs=[html_output],

            outputs=[preview_output]

        )

        refresh_preview.click(

            show_preview,

            inputs=[html_output],

            outputs=[preview_output]

        )
```

Tab 3: Color Palettes

```
 with gr.TabItem("Colors"):

        gr.Markdown("### Color Palettes")

        with gr.Row():

            style_selector = gr.Dropdown(

                ["modern", "vibrant", "ocean", "forest", "sunset", "luxury", "creative", "minimal"],

                value="modern",

                label="Select Style",

                interactive=True

            )
```

```
        palette_btn = gr.Button("Show Colors", size="lg", variant="primary")

    palette_display = gr.HTML()

    palette_btn.click(

        get_color_suggestions,

        inputs=[style_selector],

        outputs=[palette_display]

    )
```

Tab 4: Templates

```
with gr.TabItem("Templates"):
    gr.Markdown("### Quick Templates")
    # Basic Templates
    gr.Markdown("#### Basic")
    with gr.Row(elem_classes="template-grid"):
        with gr.Column():
            with gr.Group(elem_classes="template-card"):
                gr.Markdown("**📸 Portfolio**")
                gr.Markdown("*Hero, Gallery, About, Contact*")
                portfolio_btn = gr.Button("Use Template", size="sm", variant="secondary")
        with gr.Column():
            with gr.Group(elem_classes="template-card"):
                gr.Markdown("**🚀 SaaS**")
                gr.Markdown("*Hero, Features, Pricing, CTA*")
                saas_btn = gr.Button("Use Template", size="sm", variant="secondary")
        with gr.Column():
            with gr.Group(elem_classes="template-card"):
                gr.Markdown("**🏢 Company**")
                gr.Markdown("*Hero, Services, Team, Contact*")
                company_btn = gr.Button("Use Template", size="sm", variant="secondary")
    # E-Commerce & Blog
    gr.Markdown("#### E-Commerce & Blog")
    with gr.Row(elem_classes="template-grid"):
        with gr.Column():
            with gr.Group(elem_classes="template-card"):
                gr.Markdown("**🛒 Store**")
                gr.Markdown("*Hero, Products, Categories*")
                ecommerce_btn = gr.Button("Use Template", size="sm", variant="secondary")
        with gr.Column():
            with gr.Group(elem_classes="template-card"):
                gr.Markdown("**📝 Blog**")
                gr.Markdown("*Author, Posts, Categories*")
                blog_btn = gr.Button("Use Template", size="sm", variant="secondary")
        with gr.Column():
            with gr.Group(elem_classes="template-card"):
                gr.Markdown("**🍽️ Restaurant**")
```

```
                gr.Markdown("*Menu, Chef, Reservations*")
                restaurant_btn = gr.Button("Use Template", size="sm", variant="secondary")
        # Creative & App
        gr.Markdown("#### Creative & App")
        with gr.Row(elem_classes="template-grid"):
            with gr.Column():
                with gr.Group(elem_classes="template-card"):
                    gr.Markdown("**🏢 Agency**")
                    gr.Markdown("*Hero, Services, Case Studies*")
                    agency_btn = gr.Button("Use Template", size="sm", variant="secondary")
            with gr.Column():
                with gr.Group(elem_classes="template-card"):
                    gr.Markdown("**📱 App**")
                    gr.Markdown("*Screenshot, Features, Download*")
                    app_btn = gr.Button("Use Template", size="sm", variant="secondary")
            with gr.Column():
                with gr.Group(elem_classes="template-card"):
                    gr.Markdown("**💪 Fitness**")
                    gr.Markdown("*Schedule, Trainers, Plans*")
                    fitness_btn = gr.Button("Use Template", size="sm", variant="secondary")
# Connect template buttons
        portfolio_btn.click(load_portfolio, outputs=[page_title, page_description, color_style])
        saas_btn.click(load_saas, outputs=[page_title, page_description, color_style])
        company_btn.click(load_company, outputs=[page_title, page_description, color_style])
        ecommerce_btn.click(load_ecommerce, outputs=[page_title, page_description,
color_style])
        blog_btn.click(load_blog, outputs=[page_title, page_description, color_style])
        restaurant_btn.click(load_restaurant, outputs=[page_title, page_description, color_style])
        agency_btn.click(load_agency, outputs=[page_title, page_description, color_style])
        app_btn.click(load_app_landing, outputs=[page_title, page_description, color_style])
        fitness_btn.click(load_fitness, outputs=[page_title, page_description, color_style])
```

Tab 5: Guide

```
with gr.TabItem("Guide"):

        gr.Markdown("""

        # Quick Guide

        ## How to Use

        1. **Generator** - Enter title and description

        2. **Preview** - See your page live

        3. **Colors** - Browse color schemes

        4. **Templates** - Start with pre-built layouts

        ## Keywords
```

- hero, banner → Hero Section

- features, services → Features Grid

- about, team → About Section

- portfolio, gallery → Portfolio

- testimonials, reviews → Testimonials

- pricing, plans → Pricing Plans

- contact, form → Contact Form

- footer → Footer

Color Styles

- **Modern** - Professional blues

- **Vibrant** - Bold greens

- **Ocean** - Calm blues

- **Forest** - Natural greens

- **Sunset** - Warm oranges

- **Luxury** - Elegant gold/black

- **Creative** - Purple gradients

- **Minimal** - Clean grays

""")

MILESTONE 4: Deployment

Activity 4.1: Google Colab Deployment

Launch the application in Google Colab:

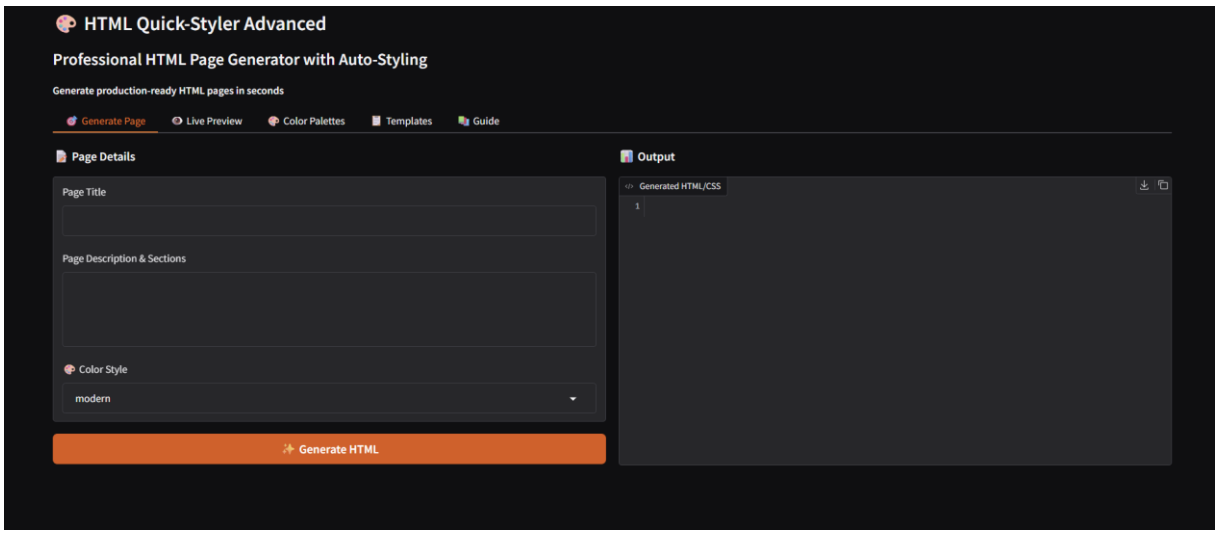
```
print("\n✅ HTML Quick-Styler Advanced Ready!\n")
demo.launch(share=True, show_error=True, show_api=False)
```

```
🚀 Loading HTML Quick-Styler Advanced...
ut actions
L Quick-Styler Advanced Ready!

/tmp/ipython-input-1159347522.py:956: DeprecationWarning: The 'show_api' parameter in launch() will be removed
  demo.launch(share=True, show_error=True, show_api=False)
Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://87205c7f5a55c5adc8.gradio.live
```

The application will be accessible via:

- Public URL: : https://87205c7f5a55c5adc8.gradio.live



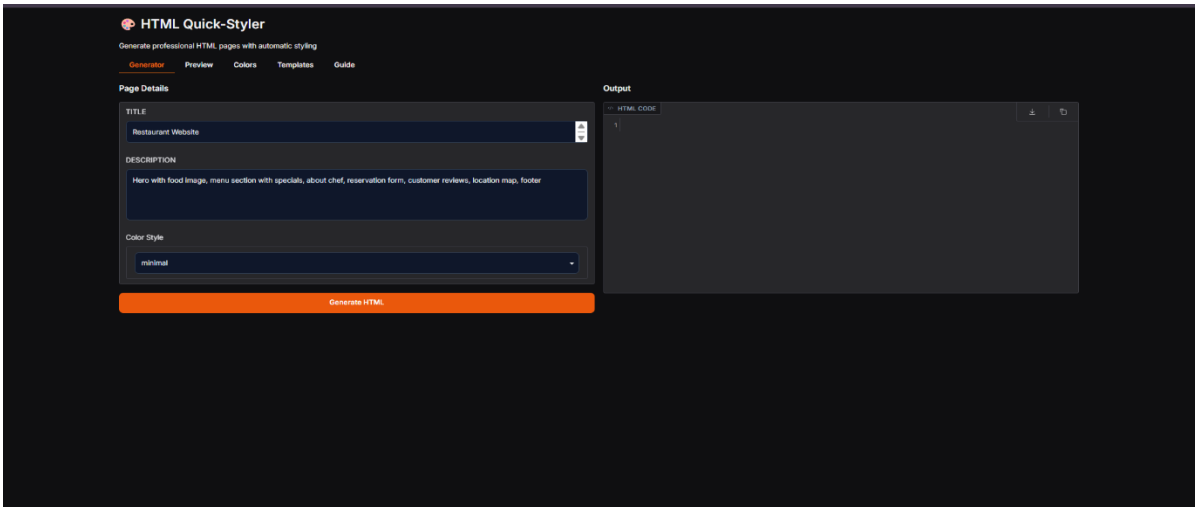
MILESTONE 5: Testing & Validation

Activity 5.1: Functional Testing

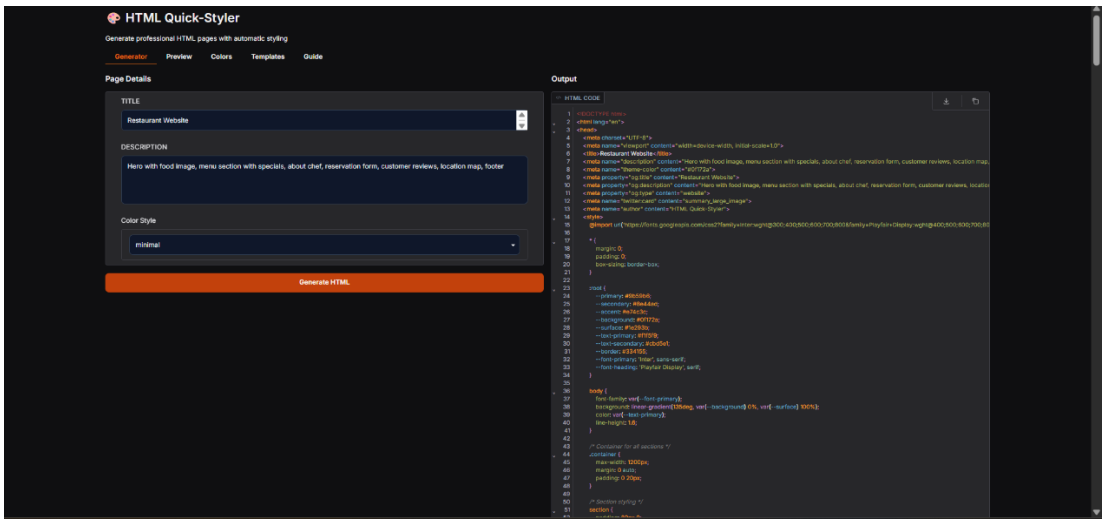
Test Case 1: Restaurant Website Generation

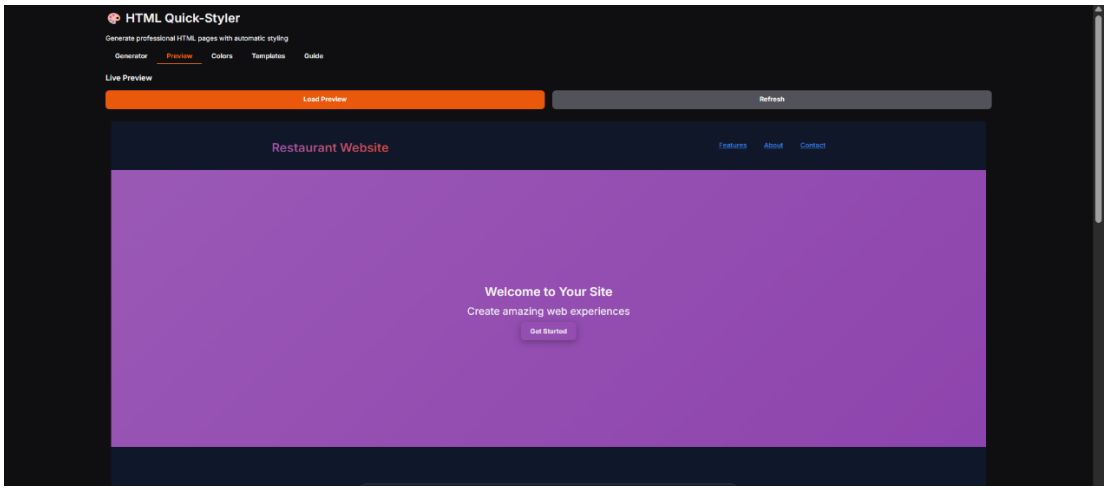
Input:

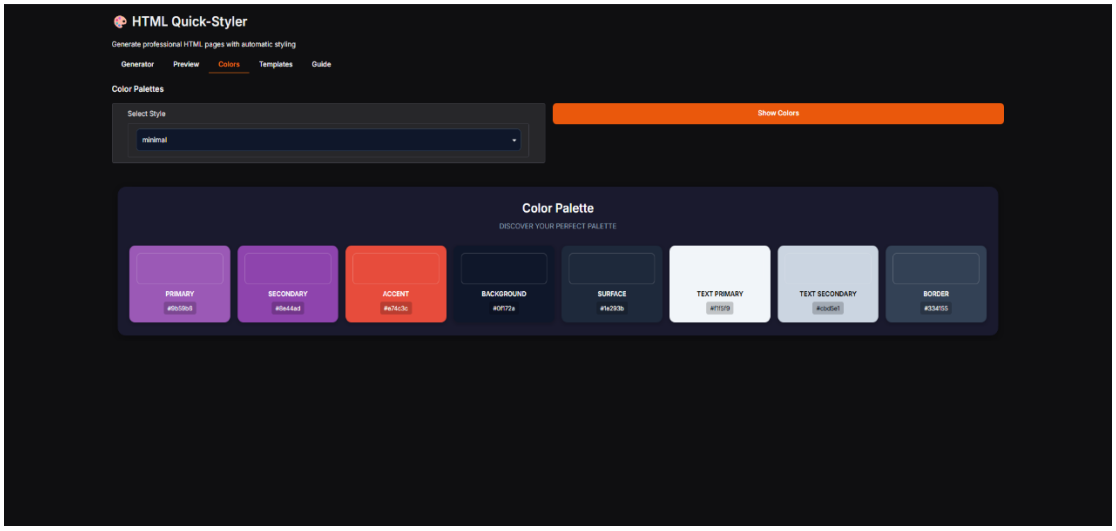
- Title: Restaurant Website
- Description: Hero with food image, menu section with specials, about chef, reservation form, customer reviews, location map, footer
- Color Style: minimal

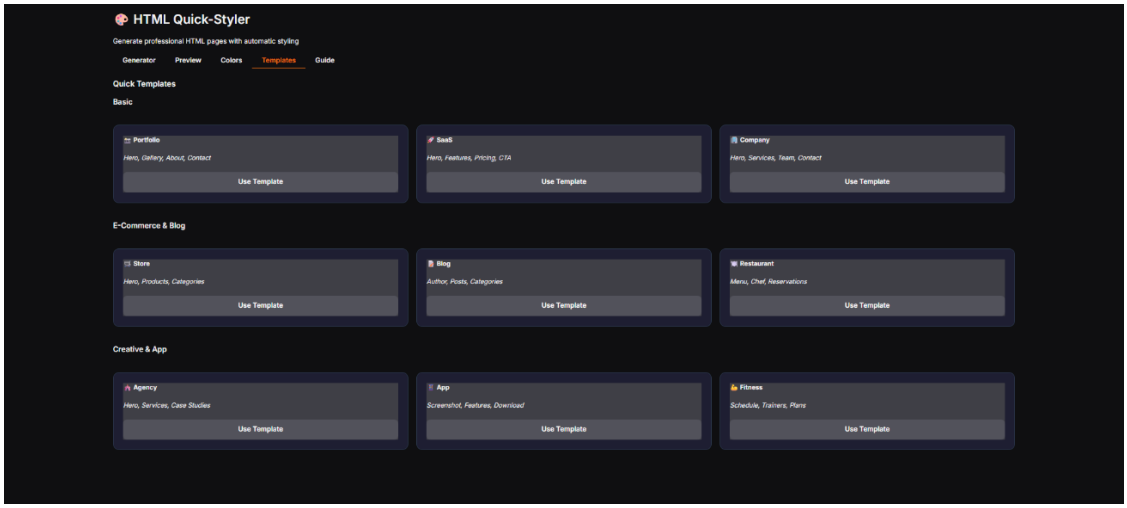


Expected Output:






HTML Quick-Styler
Generate professional HTML pages with automatic styling
Generator
Preview
Colors
Templates
Guide
Color Palettes
Select Style
minimal
Show Colors
Color Palette
DISCOVER YOUR PERFECT PALETTE
PRIMARY
#9b58b6
SECONDARY
#8e44ad
ACCENT
#e74c3c
BACKGROUND
#0f172a
SURFACE
#1e293b
TEXT PRIMARY
#f1f5f9
TEXT SECONDARY
#cbd5e1
BORDER
#334155


HTML Quick-Styler
Generate professional HTML pages with automatic styling
Generator
Preview
Colors
Templates
Guide
Quick Templates
Basic
Portfolio
Hero, Gallery, About, Contact
Use Template
SaaS
Hero, Features, Pricing, CTA
Use Template
Company
Hero, Services, Team, Contact
Use Template
E-Commerce & Blog
Store
Hero, Products, Categories
Use Template
Blog
Author, Posts, Categories
Use Template
Restaurant
Menu, Chef, Reservations
Use Template
Creative & App
Agency
Hero, Services, Case Studies
Use Template
App
Screenshot, Features, Download
Use Template
Fitness
Schedule, Trainers, Plans
Use Template

HTML Quick-Styler
Generate professional HTML pages with automatic styling
Generator
Preview
Colors
Templates
Guide
Quick Guide
How to Use
1. Generator - Enter title and description
2. Preview - See your page live
3. Colors - Browse color schemes
4. Templates - Start with pre-built layouts
Keywords
hero, banner → Hero Section
features, services → Features Grid
about, team → About Section
portfolio, gallery → Portfolio
testimonials, reviews → Testimonials
pricing, plans → Pricing Plans
contact, form → Contact Form
footer → Footer
Color Styles
Modern - Professional blues
Vibrant - Bold greens
Ocean - Calm blues
Forest - Natural greens
Sunset - Warm oranges
Luxury - Elegant gold/black
Creative - Purple gradients
Minimal - Clean grays

Conclusion

HTML Quick-Styler successfully delivers an intelligent web page generation assistant that combines intuitive design templates with IBM Granite 3.3 2B Instruct model capabilities. By leveraging Gradio's powerful interface framework and Google Colab's cloud infrastructure, the system provides fast, professional, and customizable HTML/CSS generation accessible to designers, developers, and non-technical users alike.

The platform demonstrates the viability of AI-assisted web development, making professional page creation accessible while maintaining high design standards, responsive layouts, and production-ready code quality. With eight distinct color palettes, intelligent section detection, and comprehensive template options, HTML Quick-Styler empowers users to create beautiful websites in minutes rather than hours.

Future enhancements could include custom color picker integration, JavaScript interactivity generation, component library expansion, and integration with popular web frameworks. The modular architecture allows for easy extension and customization to meet evolving user needs and design trends.